
Capacity Without Coordinate Stasis

A development study of the OLMo 32B J-space lineage

Isolated OLMo parallel phase — August 2026

Central finding

Across the architecture-matched OLMo lineage, the measured sparse-capacity estimand is stable from Base to the first released Think endpoint, while J-mapped token rows and selected spans reorganize sharply at that transition. The later 3.0-to-3.1 Think transition is geometrically much smaller. This licenses a matched-lineage association between post-training and coordinate/ selection reorganization without large measured capacity growth. It does not identify which training stage caused the change.

The decisive externalization experiment did not open

Both OLMo-3.1 endpoints passed their individual Bank-W capability gates, but the OLMo/Qwen intersection contained 16 capable families rather than the 20 required prospectively. No Bank-W intervention cell was run. External-state substitution remains a motivated hypothesis, not a resolved mechanism.

Branch:	<code>interp_jspace_olmo_lineage</code>
Scientific import boundary:	<code>3b041735d8b842de46a9c0a474fccd0c44e0841a</code>
Native tiers:	development and methods only
Registry state:	23 live events, 88 verified outputs
Latest evidence:	<code>ol-independent-reconstruction-v1</code>

This manuscript is run-specific. It does not modify the shared Phase 4 or Gemma papers and is intended for later Phase 5 integration.

Abstract

The phrase “workspace formation” conflates coordinate availability, sparse capacity, causal utilization, downstream consumption, external-state substitution, and temporal organization. We separate those axes in four architecture-matched OLMo 32B checkpoints: Base, OLMo-3 Think, OLMo-3.1 Think, and an OLMo-3.1 Instruct sibling. On a frozen 120-prompt activation corpus, the Base-to-3.0 Think equal-layer capacity contrast is +0.0154 percentage points (paired 90% interval $[-0.0121, +0.0434]$ points) with no occupancy change and is stable under a prospective equivalence rule. Same-corpus geometry is not stationary: Base-to-3.0 mapped-row movement is 0.32556 versus 0.00604 for 3.0-to-3.1, and selected-ID Jaccard is 0.333 at the three assay layers versus 1.0 on the later Think edge. A frozen router labels this a *dictionary-formation pattern*. The planned Bank-W mechanism grid is gated out because strict joint capable-family support is 16/20; no externalization outcome is inferred. Official 32B SFT and DPO stages are available for a later bounded wedge, while the current evidence cannot identify a crossed activation/lens/readout causal estimand. An independent reconstruction reproduces all headline tables, all five PNGs byte-for-byte, all fourteen exact model shards, and one frozen eight-candidate model row with zero log-probability drift. All results remain development or methods tier.

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1 Question and release boundary

Prior J-space work observed large checkpoint differences in downstream causal use, especially on the OLMo Think path. That observation admits at least six non-equivalent explanations:

1. **coordinate availability**: whether a thin verbalizable mapping is measurable at all;
2. **sparse capacity**: how many directions are recruited and how much centered energy they explain beyond matched random dictionaries;
3. **causal utilization**: whether selected content is necessary downstream beyond its geometric dose;
4. **downstream consumption**: which receivers read or transform it;
5. **external-state substitution**: whether supplied or redundant state reduces reliance on the internal channel;
6. **temporal organization**: which post-training stage installs a measurable change.

The OLMo parallel phase was designed around an O1–O2–O4 spine: first service the main Phase 4 Bank-W baseline gate, then fill the missing Base capacity cell, then use a fact-paired factorial to distinguish externalization accounts. O3 audits whether coordinates are comparable; O5 asks whether existing data identify activation, transport, and readout factors. Crucially, a failed gate is a result. It does not authorize a smaller post-hoc cohort.

Evidence status

The workstream starts from scientific boundary 3b041735d8b842de46a9c0a474fccd0c44e0841a and writes only the isolated `interpretability/jspace_olmo_lineage` namespace and Drive run root. Native IDs begin `ol-`; native tiers are development or methods. The main Phase 4 and Gemma branches remain concurrent and separate.

2 Design and estimands

2.1 Matched lineage, not randomized training

Checkpoint	Role	Inference boundary
OLMo-3 Base	pretrained anchor	first symmetric capacity cell; common-frame reference, not expected to pass every task gate
OLMo-3 32B Think	first released Think end-point	localizes the first observed transition but bundles unobserved post-training stages
OLMo-3.1 32B Think	later Think endpoint	within-path contrast after the first released Think state
OLMo-3.1 32B Instruct	sibling endpoint	valuable architecture/base match; never a fourth temporal Think point

Table 1: The model graph. Architecture is held fixed enough for a matched lineage comparison, but objective, data, formatting, and checkpoint selection are not randomized.

The allowed temporal graph is

$$\text{Base} \longrightarrow \text{3.0 Think} \longrightarrow \text{3.1 Think}, \quad \text{Base} \longrightarrow \text{3.1 Instruct (sibling)}.$$

Accordingly, “training-associated” is licensed; “caused by the Think objective” is not.

2.2 Bank-W capability and service estimand

Each OLMo-3.1 endpoint receives 24 families $\times$ 8 seeds $\times$ two load levels in the derived/once cell, for 384 unique rows. All eight answer aliases are scored by summed conditional sequence log probability. A model passes if both aggregate accuracies are at least 0.70 and the family-bootstrap 90% interval for high-minus-low accuracy lies within $[-0.08, +0.08]$. The intervention service set additionally requires at least 20 families capable for OLMo-3.1 Think, OLMo-3.1 Instruct, and the imported Qwen reference.

2.3 Symmetric capacity estimand

For every model and layer $\ell \in \{24, 32, 40\}$, activations from the identical ordered 120-prompt corpus are globally centered before nonnegative sparse pursuit. Marginal J-dictionary gain is compared with the median marginal gain of three matched random dictionaries. The lower-median persistent crossing defines occupancy K^* , and centered excess is

$$E_{J,\ell} = \left(1 - \frac{\sum_i e_{i,K^*}^J}{\sum_i e_{i,0}^J}\right) - \frac{1}{R} \sum_{r=1}^R \left(1 - \frac{\sum_i e_{i,K^*}^{(r)}}{\sum_i e_{i,0}^J}\right).$$

Raw-target pursuit is retained only as a sensitivity. Joint contrasts use 2,000 shared domain-stratified prompt-bootstrap draws. A difference is stable when its magnitude is below 0.25 percentage points, occupancy is unchanged, and the paired interval lies within the equivalence margin.

2.4 Geometry router

The geometry audit compares all six checkpoint pairs across all 21 imported source layers. It separately records operator cosine, mapped token-row similarity, stable-token CKA/neighborhood overlap, raw readout similarity, and selected-ID/span continuity at the capacity layers. A formation pattern requires Base-to-3.0 mapped movement to exceed later Think movement by both a 1.5 ratio and 0.03 absolute margin. These are descriptive thresholds frozen before aggregation; they are not intervention estimates.

3 O1: capability passes, service fails

Model	Rows	Low	High	High-low	90% CI	Capable
OLMo-3.1 Think	384	.7135	.7188	+.0052	$[-.0313, .0417]$	17/24
OLMo-3.1 Instruct	384	.7396	.7188	-.0208	$[-.0573, .0208]$	17/24
Qwen, imported	384	.8333	.8333	0	$\approx [-.0208, .0208]$	20/24

Table 2: Frozen Bank-W baseline capability results. Both OLMo models pass independently. Evidence: [ol-bank-w-capability-olmo31-think-dev-v1], [ol-bank-w-capability-olmo31-instruct-dev-v1].

The exact three-model capable-family intersection contains 16 families, not the required 20. The joint event therefore sets `olmo_phase4_service_ready=false`. The early import bundle records the full baseline evidence and `interventions_opened=false` [ol-bank-w-capability-joint-dev-v1] [ol-phase4-early-import-bundle-v1].

What 16/20 means

It does not mean that the OLMo endpoints lack baseline capability in aggregate. It means the common, model-complete family population required for the frozen factorial is too small. No model or family can

be dropped after observing this gate, and no O4 account can be selected from a grid that never opened.

4 O2: no material measured capacity growth

Checkpoint	Layer 24	Layer 32	Layer 40
Base	−0.0137 pp (2)	+0.3151 pp (2)	+0.5896 pp (2)
3.0 Think	+0.0650 pp (2)	+0.3391 pp (2)	+0.5331 pp (2)
3.1 Think	+0.0625 pp (2)	+0.3275 pp (2)	+0.5192 pp (2)
3.1 Instruct	−0.0157 pp (2)	+0.2615 pp (2)	+0.4716 pp (2)

Table 3: Own-frame centered excess in percentage points, with lower-median occupancy in parentheses. All values use the same corpus and estimator. Evidence: four `ol-capacity-*-dev-v1` model events.

The preregistered Base-to-3.0 Think equal-layer difference is 0.0001538 (+0.0154 percentage points), with paired 90% interval $[-0.0001211, 0.0004345]$ and occupancy difference exactly zero. All three primary layer rows are stable. All twelve equal-layer pair/frame summaries are stable; 46/48 classified rows are stable, while two Base-common layer-40 sensitivities narrowly cross the equivalence edge and remain unresolved.

The frozen joint verdict is `broadly_conserved_capacity_recruitment_consistent` [`ol-capacity-joint-dev-v1`]. This rejects a large capacity-growth account under the measured estimator. It does not prove equality, conserved token coordinates, or downstream causal recruitment.

5 O3: capacity stability does not imply coordinate stasis

5.1 Comparability before geometry

All six lens pairs are classified `EXACT_SAME_RECIPE_CORPUS`: same fitting texts and order, slice partition, merge inputs, hyperparameters, and selected token population. The three post-trained tokenizers produce identical token sequences on all 120 fit texts; Base differs only in the explicitly modeled BOS interface. The frozen decision is to use the existing lenses, not refit after seeing outcomes [`ol-lens-provenance-audit-v1`].

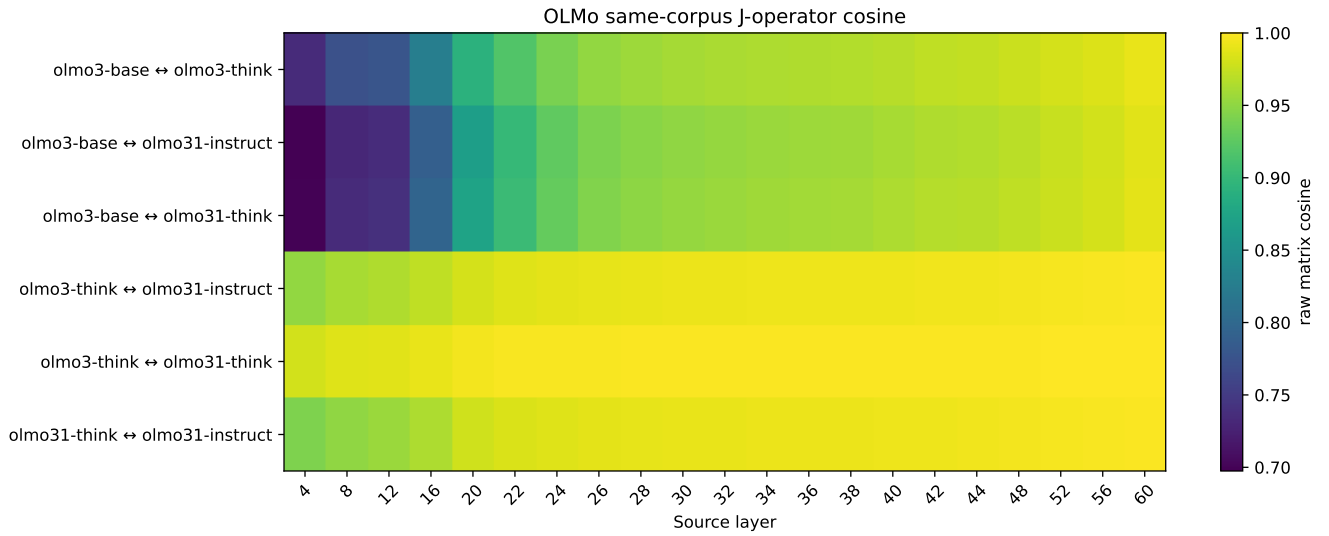


Figure 1: All-layer raw J-operator cosine for every checkpoint pair. Coarse operator continuity is high overall but has localized early-layer movement. Evidence: [ol-geometry-figures-dev-v1].

5.2 Mapped token rows move at the first Think transition

Base-to-3.0 raw-operator cosine has all-layer median 0.9614 and minimum 0.7343 at layer 4. Yet mapped token-row median cosine is only 0.6744, and the median layer-level fifth percentile is 0.5950. By contrast, raw unembedding rows have median cosine 0.9853. The discrepancy therefore lies primarily in the fitted J mapping and its token geometry rather than gross output-weight movement.

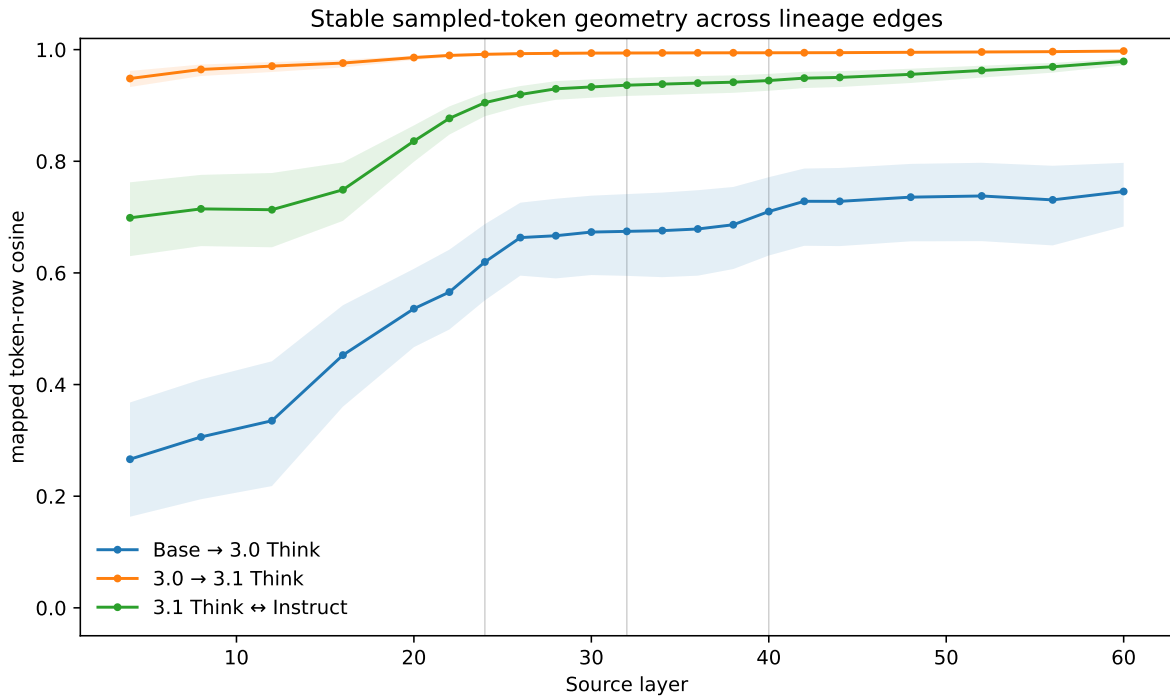


Figure 2: Mapped token-row cosine across source layers. Base-to-3.0 movement is large, while 3.0-to-3.1 is nearly stationary. The Instruct series is a sibling comparison, not a temporal continuation.

The frozen formation contrast is 0.32556 mapped movement for Base-to-3.0 versus 0.00604 for 3.0-to-3.1.

The later Think edge has raw-operator/mapped-row medians 0.99817/0.99396 and raw-unembedding median 0.999994. Thus the large movement is localized to the first released Think transition, but the released checkpoints do not identify which unobserved stage produced it.

5.3 Selected spans reorganize despite stable capacity

At layers 24/32/40, Base-to-3.0 median selected-ID Jaccard is 0.3333 at every layer, with projector overlap 0.2657/0.3336/0.4549. The 3.0-to-3.1 selected-ID Jaccard is 1.0 at all three layers and projector overlap is 0.9845/0.9895/0.9906.

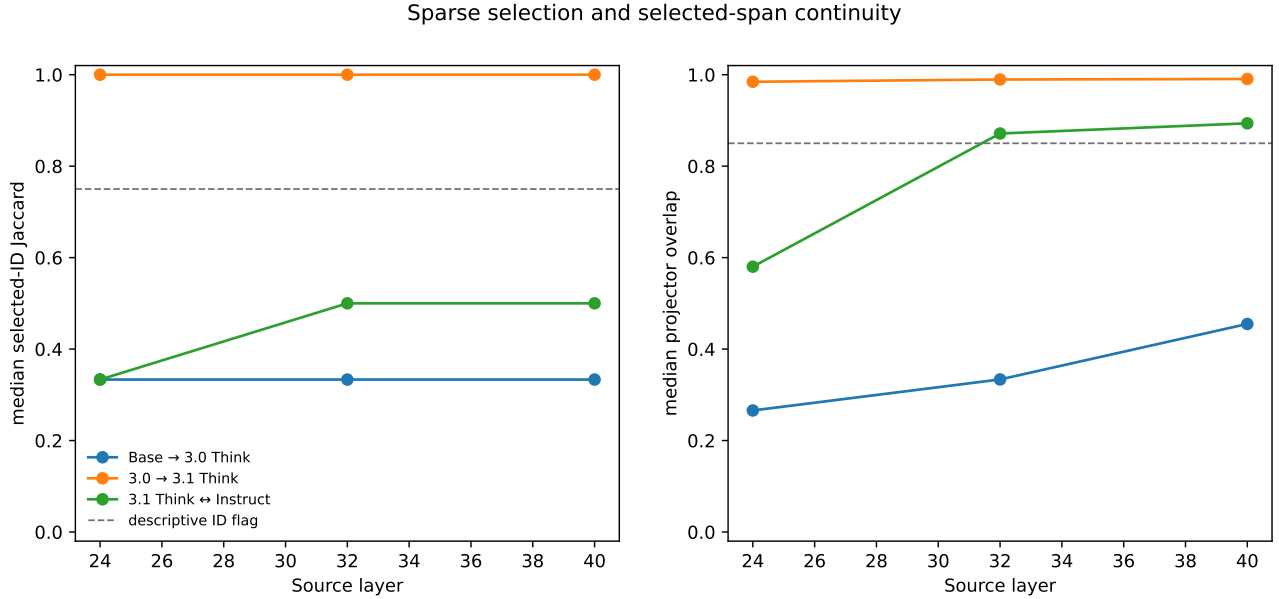


Figure 3: Selected-ID and selected-span continuity at the capacity layers. Sparse occupancy can remain stable while the identities and span of selected directions change substantially.

The complete router returns **dictionary-formation-pattern**: broad operator continuity passes, mapped-token continuity fails, selection divergence is true, the formation contrast is true, and the Instruct late-shift flag is false [ol-geometry-joint-dev-v1].

Bounded sentence-2 interpretation

Across this architecture-matched lineage, the first released Think transition is associated with strong J-mapped coordinate and selected-span reorganization without material measured sparse-capacity growth. The result is not evidence that the Think objective caused the dictionary, nor that a globally conserved coordinate system is usable at finite intervention dose.

5.4 Capacity and known causal utilization are distinct axes

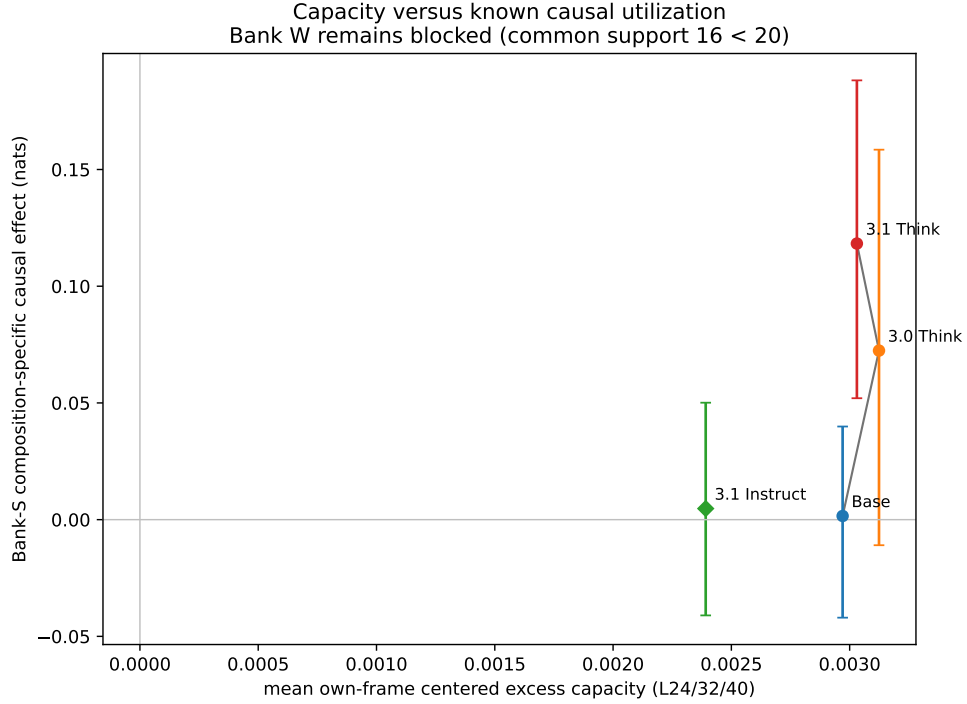


Figure 4: Own-frame capacity against previously registered Bank-S causal utilization. The causal axis is imported development evidence, not a Bank-W or O4 result. The title records the 16/20 Bank-W gate.

The state-space view is useful precisely because it resists a scalar workspace score: checkpoints with similar measured sparse capacity can differ in known causal utilization. It is descriptive triangulation. The current OLMo track did not rerun or open those Bank-S intervention outcomes.

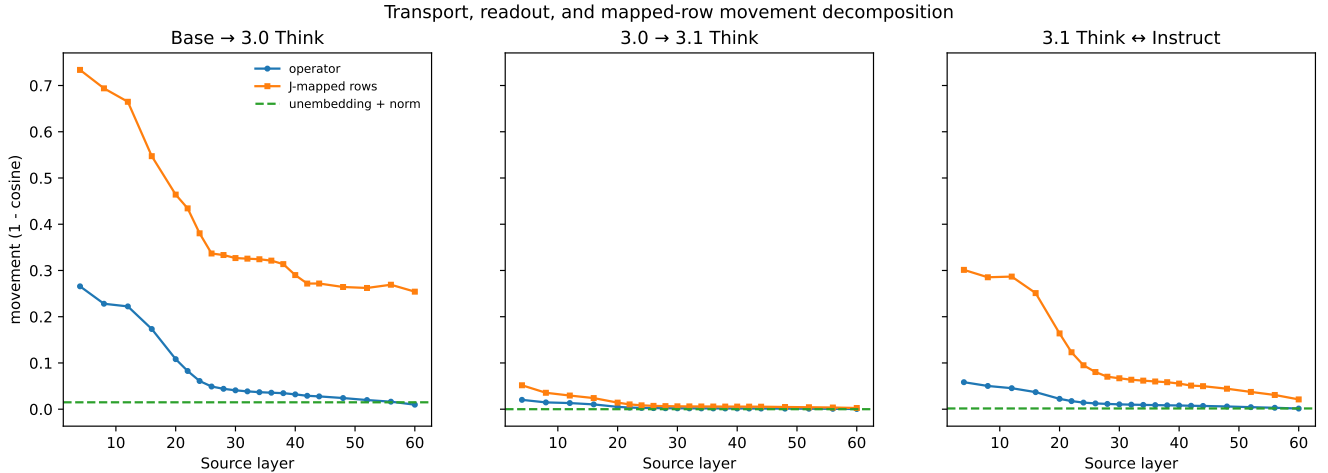


Figure 5: Movement in the fitted operator, J-mapped rows, and raw unembedding-plus-normalization readout. Structural similarity decomposes where coordinates move; it does not itself establish finite-dose causal transport.

6 Honest stopping decisions: O4, H5, and O5

6.1 O4 is gated out, not negative

The four prospective Bank-W patterns were frozen before the gate: external- state substitution, generic difficulty, capacity, and output-adjacent routing. Because common support is 16/20, no load $\times$ derivation $\times$ redundancy intervention grid exists. Consequently, conclusion-skeleton sentence 4 does not graduate.

The exact release wording is:

Existing Bank-S development evidence motivates an external-state-substitution hypothesis on the tested Think checkpoints, but the planned fact-paired Bank-W factorial was gated out at 16/20 common families; this release does not resolve an externalization mechanism.

6.2 Official intermediate stages exist, but no stage result exists

The official artifact inventory initially used byte-identical tokenizer files and conservatively returned no eligible intermediate pair. A versioned semantic audit showed that Think-SFT changes tokenizer serialization but preserves the full 100,278-entry token-ID map, normalized 100,000 BPE merges, processing components, and all 207 frozen encodings. Official 32B Think-SFT and Think-DPO artifacts are therefore eligible [ol-checkpoint-inventory-v2].

Queued, not observed

H5 is **testable-with-bounded-stage-wedge**, with SFT and DPO marked **queued-not-started**. The existing Base, 3.0 Think, and 3.1 Think checkpoints are anchors. BOS/chat-template differences and repository-only parent declarations remain qualifications. Availability is not stage attribution.

6.3 O5 is not identifiable from structural proxies

The registry has capacity and structural geometry, but no causal rows that independently cross activation model, transport lens, and output readout. It also lacks per-dictionary finite-dose transport, protected-span geometry under each readout, delivered rank/energy checks, a non-J logit-lens baseline, and a fixed common crossed-row order. The registered decision is **defer-no-identifiable-crossed-intervention-estimand** and **not-executed-no-proxy-substitution** [ol-o5-feasibility-decision-v1]. No factor estimate or null is reported.

7 Independent reconstruction

Every headline result was reconstructed outside the production aggregation path from registered sufficient statistics. The methods event checks:

- 768 Bank-W rows, all candidate-score identities, load/family accuracy, deterministic family bootstraps, and the 16/20 intersection;
- 48 capacity curve summaries, 96 prompt-bootstrap intervals, 144 joint arrays, and all 72 joint-table rows;
- 84 geometry aggregates, the complete router, six readout rows, and all protected null boundaries;
- five regenerated PNGs byte-for-byte and five valid regenerated PDFs;
- every metadata file and all fourteen exact OLMo-3.1 Think weight shards; and

- one frozen Bank-W row under the original eight-candidate batch: all eight log probabilities, predicted alias, prompt/token manifest, and -0.25 margin reproduce with maximum drift exactly zero.

The result is `[ol-independent-reconstruction-v1]`. Its JSON SHA-256 begins `e159f01d`; the payload hash begins `ee94e446`. This is computational reconstruction, not a new confirmatory cohort and not a tier upgrade.

Two rejected pre-publication attempts are part of the audit trail. The first omitted the original producer’s `no_grad` context and exhausted VRAM after all weight hashes passed. A one-candidate diagnostic fit but changed BF16 batch numerics (maximum drift 0.20499 nats; margin -0.125 instead of -0.25), so it was rejected without weakening tolerance. Restoring `no_grad` and the frozen batch reproduced every value exactly. Neither failed attempt created a registry event or durable result.

8 Claim ledger and Phase 5 router

Target	Disposition	Licensed wording
Sentence 2	narrowed; release-resolved	architecture-matched association between the first Think transition and J-mapped coordinate/ selection reorganization without material measured capacity growth; no causal training-stage attribution
Sentence 4	explicitly pending	Bank-S motivates externalization, but the Bank-W factorial was gated out at 16/20 and resolves no mechanism
H5	queue only	official SFT/DPO stages are available and semantically compatible; no stage outcome has started
O5	not identifiable	no activation/lens/readout factor estimate or null; structural proxies are not substituted

Table 4: Paper-facing claim resolution. The sentence-level source of truth is `OLMO_LINEAGE_CLAIMS_TABLE.md`.

The next integrated phase receives three bounded entries:

1. the minimal official SFT/DPO H5 wedge, with frozen interface and stage provenance checks;
2. per-checkpoint finite-dose H6 transport validation before any claim of a conserved usable coordinate system; and
3. a Bank-S-first Base/3.1-Think/3.1-Instruct pilot crossing Base versus recipient-own transport and readout, expanded only after protected-span, delivered-rank, and delivered-energy checks pass.

Bank W remains closed under version 1. Any redesigned service population is a new prospective study, not a subset repair.

9 Limitations

1. **Natural experiment.** The lineage localizes checkpoint associations but does not randomize training data, objective, or stage.
2. **Measured capacity.** Stability applies to the frozen corpus, layers, dictionary, persistence rule, random controls, and equivalence margin. It is not a theorem about all internal capacity.
3. **Transport.** Operator and mapped-row geometry do not replace per-checkpoint finite-dose causal transport.

4. **Externalization.** O4 is absent because the service population failed prospectively. Existing Bank-S evidence cannot choose the Bank-W mechanism account.
5. **Missing margins.** Exact kth/k+1 candidate-correlation gaps, protected-span overlap under crossed readouts, and core/fringe causal dose remain null.
6. **Sibling endpoint.** Instruct is not a temporal successor and cannot be used to infer reversal along the Think path.
7. **Tier.** Reconstruction improves reliability, not evidentiary tier. No OLMo-native result is confirmatory or replication evidence.

Prohibited conclusions

This release does not license “Think training creates a global workspace,” “the reasoning objective is the causal variable,” “reasoning post-training installs external-state substitution,” “the dictionaries are identical,” or “Instruct lacks a verbalizable channel.”

10 Conclusion

The OLMo lineage closes the most important capacity hole in the earlier trajectory. Base does not show a materially smaller value under the symmetric sparse-capacity estimator, yet the first released Think transition reorganizes J-mapped token rows and selected spans far more than the later Think transition. This combination favors a careful description: capacity is stable under the measured assay while coordinate/selection organization is not.

The study is equally informative where it stops. A 16/20 prospective service failure prevents the decisive Bank-W externalization grid; official SFT/DPO stages are available but unrun; and structural geometry cannot stand in for a crossed causal decomposition. These boundaries yield a sharper Phase 5 plan than an overextended mechanism claim would: test stage localization, finite- dose transport, and a minimal crossed Bank-S pilot under frozen controls.

A Evidence and artifact index

Evidence	Tier	Role
ol-foundation-v1	methods	isolation, imports, environment, preregistration
ol-bank-w-capability-joint-dev-v1	development	exact 16/20 service decision
ol-phase4-early-import-bundle-v1	methods	early hash-pinned Phase 4 capability handoff
ol-lens-provenance-audit-v1	methods	exact same-recipe/corpus classification
ol-capacity-joint-dev-v1	development	paired symmetric capacity router
ol-geometry-joint-dev-v1	development	all-layer and selected- span geometry router
ol-geometry-figures-dev-v1	development	five registered figure pairs
ol-checkpoint-inventory-v2	methods	official SFT/DPO eligibility and H5 queue
ol-o5-feasibility-decision-v1	methods	no-substitution identifiability decision
ol-independent-reconstruction-v1	methods	independent tables, figures, hashes, and model sentinel

B Key immutable hashes

- Scientific boundary: 3b041735d8b842de46a9c0a474fccd0c44e0841a.
- Early handoff JSON: debb29ef67ffa8741a4971ec2b0b21340bd5b48dc5729ac12f74f78839bf4f2b.
- Capacity joint JSON: cb35a22a...d1492.
- Capacity table: b3ad02e9...e247e; paired bootstrap: c13cebed...10fa.
- Geometry joint JSON: c09d8e73...d0f4c.
- Geometry layer table: 8692836d...9097; selection table: c2826b14...73bd.
- Geometry readout table: 93fd2108...0bd9.
- Inventory v2 JSON: e5931c3f...ad9a.
- O5 decision JSON: d31d23e4...74499.
- Reconstruction JSON: e159f01d...20542; payload: ee94e446...619f.

C Operational reproducibility

The append-only Git registry is authoritative for event order; the isolated Drive root is authoritative for heavy outputs. At paper assembly, 23 live events and 88 outputs verify, and 58 package tests plus the pinned dependency lock pass. The stable restart guide is `reports/OLMO_LINEAGE_RESUME.md`; the complete scientific snapshot is `reports/OLMO_LINEAGE_STATE_OF_RECORD.md`; and the sentence-level license is `reports/OLMO_LINEAGE_CLAIMS_TABLE.md`.